

LIMPIEZA Y TRANSFORMACIÓN DE DATOS (SQL)

1. Creación de dimensión de riesgo

```
CREATE TABLE dim_riesgo (  
    risk_id INT PRIMARY KEY,  
    risk_level VARCHAR(20) NOT NULL  
);
```

```
INSERT INTO dim_riesgo (risk_id, risk_level)  
VALUES  
(1, 'Low'),  
(2, 'Medium'),  
(3, 'High');
```

2. Añadir riesgo a la tabla de hechos

```
ALTER TABLE fact_onboarding_eventos  
ADD COLUMN risk_id INT;
```

```
UPDATE fact_onboarding_eventos  
SET risk_id =  
CASE  
    WHEN random() < 0.70 THEN 1  
    WHEN random() < 0.90 THEN 2  
    ELSE 3  
END;
```

```
ALTER TABLE fact_onboarding_eventos  
ADD CONSTRAINT fk_fact_riesgo  
FOREIGN KEY (risk_id)  
REFERENCES dim_riesgo(risk_id);
```

3. Limpieza de estado_documentacion

Antes

```
OK  
ERROR
```

Después

```
Complete  
Incomplete  
Expired  
Suspicious
```

Transformación

```
UPDATE fact_onboarding_eventos  
SET estado_documentacion = 'Complete'  
WHERE estado_documentacion = 'OK';
```

```
UPDATE fact_onboarding_eventos  
SET estado_documentacion =  
CASE  
    WHEN random() < 0.50 THEN 'Incomplete'  
    WHEN random() < 0.80 THEN 'Expired'  
    ELSE 'Suspicious'  
END  
WHERE estado_documentacion = 'ERROR';
```

Validación

```
SELECT  
    estado_documentacion,  
    COUNT(*) AS total  
FROM fact_onboarding_eventos  
GROUP BY estado_documentacion  
ORDER BY total DESC;
```

4. Normalización de canal

Eliminación de atributos incorrectos

```
ALTER TABLE dim_canal
DROP COLUMN cliente_id,
DROP COLUMN nombre,
DROP COLUMN telefono,
DROP COLUMN email,
DROP COLUMN edad,
DROP COLUMN pais,
DROP COLUMN segmento;
```

Canales finales

```
DELETE FROM dim_canal;
```

```
INSERT INTO dim_canal
VALUES
(1, 'Web'),
(2, 'Mobile App'),
(3, 'Branch');
```

5. Validaciones de calidad de datos

Riesgo

```
SELECT
    r.risk_level,
    COUNT(*) AS total
FROM fact_onboarding_eventos f
JOIN dim_riesgo r
ON f.risk_id = r.risk_id
GROUP BY r.risk_level;
```

AML

```
SELECT
    aml_flag,
    COUNT(*) AS total
```

```
FROM fact_onboarding_eventos  
GROUP BY aml_flag;
```

Verificación

```
SELECT  
    tipo_verificacion,  
    COUNT(*) AS total  
FROM fact_onboarding_eventos  
GROUP BY tipo_verificacion;
```

Estado documental

```
SELECT  
    estado_documentacion,  
    COUNT(*) AS total  
FROM fact_onboarding_eventos  
GROUP BY estado_documentacion;
```

Estado evento

```
SELECT  
    estado_evento,  
    COUNT(*) AS total  
FROM fact_onboarding_eventos  
GROUP BY estado_evento;
```

Canal

```
SELECT  
    c.canal_nombre,  
    COUNT(*) AS total  
FROM fact_onboarding_eventos f  
JOIN dim_canal c  
ON f.canal_id = c.canal_id  
GROUP BY c.canal_nombre;
```